Research Statement

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My research focuses on number theory, where I solve problems in number theory using various computational tools including **Computer Algebra Systems** (SageMath, MAGMA), **Proof Assistants** (Lean), and **Machine Learning/AI**.

1 Number Theory with Computer Algebra Systems

Modular forms and optimal sphere packings The *optimal sphere packing problem* asks the densest arrangement of non-overlapping unit balls in $\mathbb{R}^d$. The problem is solved in dimensions $d = 1, 2, 3, 8$ and 24, where the last two cases are recently resolved by Maryna Viazovska and her collaborators. Their method constructs so-called *magic functions* for the Cohn–Elkies linear programming bound, using (quasi)modular forms.

One of the main steps of the proofs is the verification of certain inequalities involving (quasi)modular forms, corresponds to the linear constraints of the magic functions. The original arguments relied on numerical approximations and extensive computer assisted computations, and it is natural to ask if there is a more general and conceptual proof for these inequalities. A more direct proof is recently given by Dan Romik for the dimension 8 case.

In [12], I have found *algebraic* proofs for both the 8- and 24-dimensional cases that avoid reliance on numerical approximations, by developing a simple yet effective theory of *positive* and *completely positive* quasimodular forms. Additionally, I uncovered connections to Kaneko and Koike's *extremal quasimodular forms*, which are the quasimodular forms with largest possible orders at cusp of given weight and depth. They are conjectured to have non-negative Fourier coefficients, where I proved the conjecture for the case of depth 1 in *loc. cit*. A key ingredient in my approach is the use of modular differential equations satisfied by these forms. My work opens new possibilities for generalizing Viazovska's construction to dimensions beyond 8 and 24, based on Fourier eigenfunctions constructed by Feigenbaum, Grabner, and Hardin. In particular, this lead to new upper bounds for the sign uncertainty principle à la Bourgain–Clozel–Kahane in certain dimensions.

Theorem (L., In progress)**.** The optimal constant $\mathrm{A}_+(d)$ for the (+1)-sign uncertainty principle of Bourgain–Clozel–Kahane satisfies $\mathrm{A}_+(d) \leq \sqrt{2\left\lfloor \frac{d}{16} \right\rfloor + 2}$ when $d \equiv 0 \pmod 4$ and $d \leq 36000$.

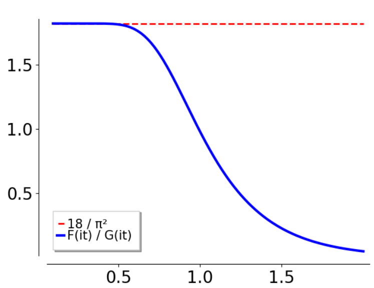

Figure 1: Plot of $F(it)/G(it)$.

These proofs were inspired by experiments using SageMath. For instance, the plot of a quotient of modular forms (Figure 1) suggested key properties - monotonicity and the limiting behavior as $t \to 0^+$ - to prove the inequality $F(it) < \frac{18}{\pi^2} G(it)$ for dimension 8. Likewise, the quasimodular identities in [12] can be verified both computationally and by hand.

Based on a similar idea, I also studied other modular form inequalities appear in the proof of the universal optimality of the E_8 and Leech lattice. In particular, I gave an algebraic proof of one of such inequalities, by relating it to extremal quasimodular forms again [13]. Not only proving the inequality, I propose general approach to prove monotonicity of the functions of the form $t \mapsto t^m F(it)$ for a quasimodular form F.

Currently, I am extending this theory to higher-level quasimodular forms, aiming to prove analogous conjectures for extremal forms of level ≥ 2, which are studied by Sakai and Tsutsumi. I am also working on improving Cohn and Trantafillou's *dual* bounds for the Cohn–Elkies bound, by constructing new summation formulae from other types of automorphic forms.

Function field arithmetic In the summer of 2025, I mentored four students at the Berkeley Mathematics REU on projects in the arithmetic of function fields. We explored analogues of classical Diophantine problems and prime number races [4, 5]. For example, we completely characterized all perfect powers in the sequence of Fibonacci polynomials over finite fields, showing that infinitely many occur - in sharp contrast to the classical integer case, where only finitely many exist. We also discovered infinite families of *ties* in function field prime races, proving that for certain moduli m and residue classes a, b, the counts $\pi(a, m, N)$ and $\pi(b, m, N)$ of monic irreducible polynomials of degree N are equal for infinitely many N. These results were established both via explicit formulas involving L-functions and by constructing bijections using $\mathrm{GL}_2(\mathbb{F}_q)$-actions.

Separately, I studied a function field analogue of Shanks' higher order Chebyshev's bias [14]. Shanks conjectured biases in the sign of $\lambda(n)\chi_{-4}(n)$, where λ is the Liouville function and χ_{-4} is the quadratic character mod 4. I studied the analogous problem for polynomials over $\mathbb{F}_q$, comparing the number of monic polynomials f satisfying $\lambda(f)\chi(f) = 1$ or $\lambda(f)\chi(f) = -1$, where $\chi = \chi_m$ is a quadratic Dirichlet character modulo square-free polynomial m. Especially, I proved the following theorem:

Theorem (L. [14])**.** Let $A^{\lambda}_{\pm}(n; m)$ be the number of non-constant monic polynomials $f \in \mathbb{F}_q[T]$ of degree at most n with $\lambda(f)\chi_m(f) = \pm 1$. Assume that Grand Simplicity Hypothesis holds for the Dirichlet L-function $L(s, \chi_m)$. Then we have

$$\lim_{n\to\infty} \frac{\#\{1 \leq k \leq n : A^{\lambda}_{+}(k; m) > A^{\lambda}_{-}(k; m)\}}{n} > \frac{1}{2}$$

where a closed formula for the density is obtained. When $q \equiv 1 \pmod 4$, the density can be made arbitrarily close to $\frac{1}{2}$ by choosing m appropriately with large enough degree.

This answers a function field analogue of the 22nd problem in the *Comparative Prime Number Theory Problem List* by Hamieh et al. I am also working on other problems including the average degree of least degree k-th power nonresidue modulo irreducible primes and the average number of divisors of polynomials over function fields, questions whose integer analogues are either classical or still open.

Maass wave forms, Quantum modular forms, and Hecke operators In my undergraduate and master's theses, I studied Henri Cohen's Maass wave form, an explicit form constructed from a Ramanujan q-series whose coefficients are related to a Hecke character of $\mathbb{Q}(\sqrt{6})$. Although Cohen conjectured that this form is a Hecke eigenform, the usual Hecke operators do not apply because its multiplier system is not induced by a Dirichlet character. I developed Hecke operators for more general multiplier systems and proved Cohen's conjecture using these operators [10, 11]. I also showed that the Lewis–Zagier period construction, which associates quantum modular forms to Maass wave forms, is Hecke-equivariant, yielding nontrivial identities relating values at p-th roots of unity to the p-th Fourier coefficients of the Maass wave form. The same framework applies to the Maass wave form constructed by Li, Ngo, and Rhoades.

2 Number Theory with Proof Assistant (Lean)

Formalization of 8-dimensional optimal sphere packing I serve as one of the maintainers of a project to formalize in Lean 4 Viazovska's proof that the E_8 lattice yields the optimal sphere packing in $\mathbb{R}^8$. The project is led by Sidharth Hariharan and also includes Chris Birkbeck, Bhavik Mehta, and Maryna Viazovska, with many contributors from Lean community. The project also incorporates my algebraic proof of the modular form inequalities described above. It allow us to avoid formalizing interval arithmetic and certain deep analytic results from number theory, such as bounds for the Fourier coefficients of meromorphic modular forms derived from estimates for Kloosterman sums. Thus far, we have completed the formalization of the E_8 lattice and its relevant properties, the differential theory of modular and quasimodular forms required for the proof, and the necessary modular form inequalities, and some of them already upstreamed to `mathlib` (for example, see PR #37979 and PR #36963). We are currently formalizing Viazovska's magic function and relating them to the modular form inequalities. The project is publicly available at `Sphere-Packing-Lean`, and a detailed progress report appears in [7].

Formalization of q-series In joint work with Kenny Lau and Ken Ono, we developed a formalization of the theory of q-series in Lean 4 [8]. Our framework includes foundational constructions such as q-Pochhammer symbols and q-binomial coefficients, as well as q-hypergeometric series, Bailey's lemma, and numerous related identities. In particular, we formalized the Jacobi triple product identity, Euler's pentagonal number identity, and the Rogers–Ramanujan identities, which are among the most celebrated identities in the subject. Although versions of these identities had previously been formalized in Lean and Isabelle/HOL, our aim was to identify a general class of rings over which they remain valid. To this end, we introduced the notion of a *strongly non-archimedean ring*; a topological ring admitting a neighborhood basis at zero consisting of subsets closed under addition and multiplication. This framework encompasses the settings used in previous formalizations, including formal power series rings over $\mathbb{Z}$. This work was carried out during my internship at Axiom Math in spring 2026.

Lean-GAP: A Dataset of Formalized Graduate Algebra Problems High-quality datasets of paired informal and formal mathematical statements are essential for advancing autoformalization. Despite recent improvements, a syntactically valid formalization need not faithfully represent the original natural-language statement, so careful human verification remains necessary. In joint work with Byung-Hak Hwang, Hyojae Lim, Jihoon Hyun, Ilkyoo Choi, Yeachan Park, Jineon Baek, Hyukpyo Hong, Keewoo Lee, Jaeseong Heo, Hyungryul Baik, Chul-hee Lee, and Kyu-Hwan Lee, we formalized in Lean the *statements* of 430 exercises from Dummit and Foote's *Abstract Algebra*, approximately 22% of its 1,966 exercises [15]. Multiple human and LLM reviewers checked each statement, identifying several errors in the initial LLM-generated formalizations. We aim to expand the collaboration and eventually formalize all 1,966 exercises.

Formalization of polynomial FLT In collaboration with Jineon Baek, we give a complete formalization of the Mason–Stothers theorem in Lean 4 [2], which implies Fermat's Last Theorem for polynomials. While earlier formalizations existed in HOL (Eberl) and Lean 3 (Wagemaker), our work provides a complete Lean 4 proof and a careful comparison with previous approaches [2, Section 7]. The entire proof is now integrated into `mathlib`, which can be found in PR #15706 and PR #18882.

Formal Conjectures project I occasionally contribute to Google DeepMindâĂŹs `Formal Conjectures` project, which aims to formalize the statements of mathematical conjectures and theorems (without proofs) in Lean 4. My contributions include conjectures in number theory such as the Ramanujan–Petersson conjecture, Lehmer's conjecture, Hall's conjecture, congruent number problem, Hilbert's 17th problem, Lam–Litt conjecture, and several of Erdös problems (#9, #307, #364, #494, #946, #961, #975).

3 Number Theory with Machine Learning

Machine Learning Galois Groups Recent work by Yang-Hui He, Kyu-Hwan Lee, and Thomas Oliver applied machine learning to study invariants of number fields, including Galois groups. While simple algorithms achieved high prediction accuracy, the underlying classification mechanisms were not well understood.

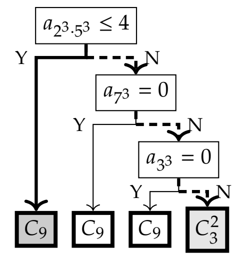

Figure 2: Decision tree classifying Galois group of nonic Galois extensions.

In a joint work with Kyu-Hwan Lee,ãĚŠ we addressed this gap by reverse-engineering the decision logic of ML models used to classify Galois groups of Galois number fields [9]. In particular, we studied the classification logic of decision trees where the Dedekind zeta coefficients are used as features, and we found that the models only use a small number of the coefficients, typically at indices that are perfect powers. For example, Figure 2 shows the decision tree predicting the Galois groups of nonic Galois extensions (which obtained 100% accuracy on a test set) from zeta coefficients up to $a_{1000}(K)$, where the model only uses three coefficients: a_{1000}, a_{343}, and a_{27}, which are all cubic indices. This empirical observation inspired the following theorem, previously absent from the literature:

Theorem (Lee–L.)**.** Let ℓ be a prime and $K/\mathbb{Q}$ be a Galois extension of degree ℓ^2, hence $\mathrm{Gal}(K/\mathbb{Q}) \simeq C_{\ell^2}$ or C_ℓ^2. Then $\mathrm{Gal}(K/\mathbb{Q}) \simeq C_{\ell^2}$ if and only if there exists a prime p such that $a_{p^\ell}(K) = 0$.

We also studied degree $4, 6, 8, 9, 10$ number fields, obtaining analogous results and showing that parts of the learned classification logic are, in fact, *provably correct*. Our work demonstrates how machine learning can be used as a tool for generating conjectures and aiding in the discovery of rigorous mathematical theorems.

Machine Learning Minimal Weierstrass Equations Every elliptic curve $E/\mathbb{Q}$ admits a unique *reduced minimal Weierstrass model* $E\colon\ y^2+w_1xy+w_3y = x^3+w_2x^2+w_4x+w_6$ with integral coefficients $w_1, w_2, w_3, w_4, w_6 \in \mathbb{Z}$ such that $w_1, w_3 \in \{0, 1\}, w_2 \in \{-1, 0, 1\}$. Its discriminant has the smallest possible absolute value among all integral Weierstrass models of E.

In joint work with Barinder S. Banwait, Xiaoyu Huang, Kyu-Hwan Lee, Thomas Oliver, and Alexey Pozdnyakov [3], we prove that the first three reduced minimal Weierstrass coefficients w_1, w_2, w_3 can be recovered from the Frobenius traces $a_2(E)$ and $a_3(E)$, together with the parity of the conductor $N(E)$. More precisely, we obtain the following explicit formulae

$$w_1 = [a_2(E)]_2, \quad w_2 = [a_3(E) + 1 - w_1]_3 - 1, \quad w_3 = (1 - w_1)[N(E)]_2 + w_1\,[N(E) + 1 + w_2 + (a_2(E) + 1)/2]_2\,,$$

where $[m]_n \in \{0, \dots, n-1\}$ is the least nonnegative residue of m modulo n. These formulae, which appear to be new, were first observed by applying decision trees to the elliptic curve database in the L-functions and Modular Forms Database (LMFDB).

My future goal is to scale-up the research and to discover more new results, possibly discover more efficient algorithms to compute various invariants of number fields or other number theoretic objects (such as automorphic forms and abelian varieties), inspired from machine learning.

4 Additional Projects

My interest is not restricted to computational number theory. I'm interested in various subjects, including

- Automorphic representations and Poisson summation formula
- Discrete geometry
- Homomorphic encryption and privacy-preserving machine learning

Modulation groups and Poisson summation formula In a joint work with Jayce R. Getz, Armando Gutiérrez Terradillos, Farid Hosseinijafari, Bryan Hu, Aaron Slipper, Marie-Hélène Tomé, HaoYun Yao, and Alan Zhao, we defined and studied *modulation groups* [6]. They can be viewed as generalizations of metaplectic groups acting on suitable Schwartz spaces of spherical varieties. More precisely, for an affine algebraic group H over a local field F, a "niceâĂŹâĂŹ affine H-scheme X, an H-representation V, and a H-equivariant map $\omega : X \to V$, the *modulation group* $\Psi_\omega(F)$ is defined as a subgroup of the automorphism group of the Schwartz space $\mathcal{S}(X(F))$, generated by the action of $V^\vee(F)$, $H(F)$, and *Fourier transforms*. We compute this group in several cases and show that it consists of the F-points of an algebraic group or a metaplectic group. In particular, we compute the group when ω is the identity map or the symmetric square map on $\mathbb{G}_a^n$, or when $\omega : X \hookrightarrow \mathbb{G}_a^{2n}$ is a quadratic cone.

One can also globalize the definition to obtain an adelic version of the modulation group, and we prove that the Poisson summation formula holds if and only if a theta series associated with ω gives an "automorphic representation of $\Psi_\omega(\mathbb{A}_F)$," although the precise definition of this notion is not yet settled. This suggests that modulation groups will play an important role in the Braverman–Kazhdan program, whose goal is to study automorphic L-functions and Langlands functoriality via the Poisson summation formula.

This work was initiated through the 2024 Duke Research Scholars program and supported by the Duke Number Theory RTG grant.

Conway–Soifer conjecture - homothetic case Consider an equilateral triangle with side length $n + \varepsilon$, where $n \geq 1$ is an integer and $\varepsilon > 0$ is sufficiently small. What is the minimum number of unit equilateral triangles required to cover this larger triangle? By considering the area, it is straightforward to show that at least $n^2 + 1$ unit triangles are necessary. Conway and Soifer provided two different ways to cover the large triangle using $n^2 + 2$ unit triangles and conjectured that this is the minimum number needed.

In collaboration with Jineon Baek, we proved that this conjecture holds if we restrict the covering to *homothetic* triangles, i.e., when the sides of the unit triangles are parallel to those of the large triangle (aligned either as $\triangle$ or $\triangledown$) [1]. Specifically, we established the following general result:

Theorem (Baek–L. [1])**.** A triangle is called a horizontal triangle of base b and height h if one of its sides of length b is parallel to the x-axis, and its height h is measured along the y-axis. Then $n^2 + 1$ horizontal triangles of base b and height h cannot cover a horizontal triangle of base nb and height greater than nh.

Our proof is elementary, and we also determined the largest possible values of ε such that an equilateral triangle of side length $n + \varepsilon$ can be covered by either $n^2 + 2$ or $n^2 + 3$ homothetic unit triangles. Specifically, these values are $\varepsilon = 1/(n + 1)$ and $\varepsilon = 1/n$, respectively. We believe that our method can be generalized to higher dimensions or extended to determine the largest side length of an equilateral triangle that can be covered by $n^2 + k$ homothetic unit triangles for $1 \leq k \leq 2n$.

Encrypted transfer learning with homomorphic encryption While working as a Research Engineer at CryptoLab, I developed `HEaaN.SDK`, a privacy-preserving machine learning library based on the Cheon–Kim–Kim–Song (CKKS) homomorphic encryption (HE) scheme. Although CKKS supports approximate arithmetic on encrypted real and complex numbers, encrypted computations are orders of magnitude slower than their plaintext counterparts. Practical applications therefore require algorithms to be carefully redesigned in an *HE-friendly* manner. A major challenge was multiclass classification, as most previous work focused on low-dimensional binary tasks. Extending these methods required efficient encrypted softmax evaluation for large inputs and scalable encrypted matrix multiplication.

We addressed these challenges in HETAL (**H**omomorphic **E**ncryption-based **T**ransfer **L**earning) [16]. For softmax evaluation, we combined Cheon et al.'s homomorphic comparison method with homomorphic domain extension, enabling stable max-subtraction normalization while reducing approximation errors and enlarging the admissible input range. We also developed specialized algorithms for computing $AB^\intercal$ and $A^\intercal B$ without costly transpose operations, further optimizing them through tiling and complex packing. These techniques achieved speedups of 1.8× to 323× over previous methods and enabled encrypted fine-tuning of vision and language models on five benchmark datasets within one hour using a single A40 GPU. This work was selected for an oral presentation at ICML 2023.

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